Session 03 - Menu Loop and In-Memory Documents - Tests

Python Contoso Vault - Session 03 of 07

Part 1 - Core: vault.py

Goal

Wrap the cipher in a menu that keeps running until the user quits. Let the user create documents, list them and read them back. Keep each body encrypted in memory, and give a reader three attempts to type the right key.

Run `python vault.py` in the terminal and type each input line, then press Enter. To try one function, open the Python shell in the same folder and run `from vault import check_key, create_document`.

Provided

Copy your own `caesar.py` from your Session 02 folder into this folder. It defines `encrypt(plaintext: str, key: int) -> str` and `decrypt(ciphertext: str, key: int) -> str`.

The menu is these four lines:

```
1. Create
2. List
3. Read
4. Quit
```

Instructions

1. Import `encrypt` and `decrypt` from `caesar.py`.
2. Define a function `create_document` that takes the documents as a list, a title as a string, a body as a string and a key as an integer.
   2.1. Give the document a unique ID, as an integer. IDs count up from 1 in the order the documents are created, so the third document has ID 3.
   2.2. Encrypt the body with the key.
   2.3. Encrypt `CONTOSO-VAULT` with the key, as the check value.
   2.4. Build the document as a dictionary with the keys `id`, `title`, `body` and `check`.
   2.5. Add the document to the end of the documents.
   2.6. Return the document as a dictionary.
3. Define a function `check_key` that takes a document as a dictionary and a key as an integer.
   3.1. Decrypt the document's check value with the key.
   3.2. Return `True` if the result is `CONTOSO-VAULT`, and `False` otherwise.
4. When the file runs as a program:
   4.1. Start the documents as an empty list.
   4.2. Repeat until the user quits:
      4.2.1. Print the menu.
      4.2.2. Read a choice as a string.
      4.2.3. On `1`:
         4.2.3.1. Read a title, a body and a key.
         4.2.3.2. Create the document.
         4.2.3.3. Print `Saved document N.`, where `N` is its ID.
      4.2.4. On `2`:
         4.2.4.1. Print one line `ID. Title` per document, in ID order.
         4.2.4.2. Print `No documents.` if there are none and go back to the menu.
      4.2.5. On `3`:
         4.2.5.1. Read an ID.
         4.2.5.2. Print `No such document.` and go back to the menu if no document has that ID.

4.2.5.3. Otherwise, give the user three attempts at the key. For each attempt:
4.2.5.3.1. Read a key.
4.2.5.3.2. Print the decrypted body and go back to the menu if the key is right.
4.2.5.3.3. Print `Wrong key.` if the key is wrong.
4.2.5.4. Print `Too many attempts.` and go back to the menu after the third wrong key.
4.2.6. On `4`, print `Goodbye.` and end the program.
4.2.7. On any other choice, print `Invalid choice.`

Constraints

- Read each value with a prompt on the same line, as in `input("Title: ")`. Each prompt is its label, a colon and one trailing space. The prompts are:
  - `"Choice: "` after the menu lines.
  - `"Title: "`, `"Body: "` and `"Key: "` at Create.
  - `"ID: "` then `"Key: "` for each attempt at Read.
- After `Please enter a whole number.` or `Key must be from 1 to 25.`, the prompt that asked for the value prints again.
- Type each prompt exactly as shown. A program with different prompt text does not meet the test plan.
- The menu prints before every choice, as the exact four lines above.
- Every input is one line. Create reads the title, then the body, then the key. Read reads the ID, then each key.
- A key at create must be a whole number from 1 to 25. Any other whole number prints `Key must be from 1 to 25.` and asks for the key again.
- A key or ID that is not a whole number prints `Please enter a whole number.` and asks again. It does not use an attempt.
- Two keys that differ by 26 open the same document.
- `create_document` returns the document with the ciphertext as its body.
- `create_document` and `check_key` neither print nor read input.
- Importing the file reads no input and prints nothing.
- The program never stops with a Python error message, whatever the user types. It ends only when the user chooses `"4"`.

Design

The following rules say how to write the code. Follow them.

- Decide whether a guessed key is right with `check_key`.
- Keep only the ciphertext body in memory. Never store the plaintext body.
- Give every variable and function a type hint.

Input validation

Required for every key and ID, as the constraints set out. Any title and body are valid, including empty ones.

Test plan

Function tests

`documents` starts as an empty list. `document_1` is the result of test 1.

Test	Call	Expected output	Actual output	Pass
1	`create_document(documents, "Payroll", "Salaries go out on 28 November.", 3)`	`{"id": 1, "title": "Payroll", "body": "Vdodulhv jr rxw rq 28 Qryhpehu.", "check": "FRQWRVR-YDXOW"}`. `documents` holds it		
2	Then `create_document(documents, "Kandy rent", "Kandy office rent is Rs 150,000.", 11)`	`{"id": 2, "title": "Kandy rent", "body": "Vlyoj zqqtnp cpye td Cd 150,000.", "check": "NZYEZDZ-GLFWE"}`. `documents` holds 2		
3	`create_document([], "Empty", "", 5)`	`{"id": 1, "title": "Empty", "body": "", "check": "HTSYTXT-AFZQY"}`		
4	`check_key(document_1, 3)`	`True`		
5	`check_key(document_1, 4)`	`False`		
6	`check_key(document_1, 29)`	`True`		
7	`check_key(document_1, -23)`	`True`		
8	`check_key(document_1, 0)`	`False`		

Program tests

`python vault.py` reads the input lines in order. "Create Payroll" stands for the four input lines `"1"`, `"Payroll"`, `"Salaries go out on 28 November."`, `"3"`. Every run prints the four menu lines and the prompt `"Choice: "` before each choice, and ends with the `Goodbye.` line. The expected output cells leave out the menu lines, the prompts and the `Goodbye.` line. They list the remaining output lines, such as `Saved document 1.`

Test	Input lines	Expected output	Actual output	Pass
9	`4`	Only the menu and `Goodbye.`		
10	`2`, `4`	`No documents.`		
11	Create Payroll, `2`, `4`	`Saved document 1.`, `1. Payroll`		
12	Create Payroll, `3`, `1`, `3`, `4`	`Saved document 1.`, `Salaries go out on 28 November.`		
13	Create Payroll, `3`, `1`, `5`, `3`, `4`	`Saved document 1.`, `Wrong key.`, `Salaries go out on 28 November.`		
14	Create Payroll, `3`, `1`, `5`, `6`, `7`, `2`, `4`	`Saved document 1.`, `Wrong key.` three times, `Too many attempts.`, `1. Payroll`		
15	Create Payroll, `3`, `1`, `abc`, `5`, `6`, `3`, `4`	`Saved document 1.`, `Please enter a whole number.`, `Wrong key.`, `Wrong key.`, `Salaries go out on 28 November.`		
16	`1`, `Payroll`, `Salaries go out on 28 November.`, `abc`, `0`, `26`, `3`, `4`	`Please enter a whole number.`, `Key must be from 1 to 25.`, `Key must be from 1 to 25.`, `Saved document 1.`		
17	`3`, `1`, `4`	`No such document.`		
18	Create Payroll, `3`, `two`, `0`, `4`	`Saved document 1.`, `Please enter a whole number.`, `No such document.`		
19	`7`, `x`, empty line, `4`	`Invalid choice.` three times		
20	Create Payroll, `1`, `Kandy rent`, `Kandy office rent is Rs 150,000.`, `11`, `2`, `3`, `2`, `11`, `4`	`Saved document 1.`, `Saved document 2.`, `1. Payroll`, `2. Kandy rent`, `Kandy office rent is Rs 150,000.`		
21	`import vault`	Reads no input and prints nothing		

In every program test, the program ends after `Goodbye.` and shows no Python error message.

Done criteria

- [ ] Every test plan row gives exactly the expected result.
- [ ] The script runs from the terminal with `python vault.py`.

Part 2 - Challenge: leak.py

Goal

Show why a stored check value is a weakness. Recover a document's key from its check value alone, without trying any keys.

Keep `caesar.py` and `vault.py` in the same folder as `leak.py`. Open the Python shell there, run `from leak import recover_key`, then call `recover_key` with each document named in the test plan.

Provided

`vault.py` defines `create_document(documents, title, body, key) -> dict` and `check_key(document, key) -> bool`. `caesar.py` defines `decrypt(ciphertext, key) -> str`. Import them only if `recover_key` uses them.

Instructions

1. Define a function `recover_key` that takes a stored document as a dictionary, as `create_document` returns it, with the keys `id`, `title`, `body` and `check`.
   1.1. Work out the key from the document's check value, which is `CONTOSO-VAULT` encrypted with that key. The key is an integer from 1 to 25.
   1.2. Return the key as an integer.

Constraints

- The returned key opens the document: `check_key(document, recover_key(document))` is `True`.
- The function returns its result. It neither prints nor reads input.
- Importing the file reads no input and prints nothing.

Design

The following rules say how to write the code. Follow them.

- Work out the key without trying keys.
- Call `decrypt` and `check_key` at most once per document.
- Give every variable and function a type hint.

Input validation

Not required. Assume every document comes from `create_document` with a key from 1 to 25.

Test plan

Test	Document	Expected output	Actual output	Pass
1	Check value `FRQWRVR-YDXOW`	`3`		
2	Check value `NZYEZDZ-GLFWE`	`11`		
3	Check value `DPOUPTP-WBVMU`	`1`		
4	Check value `BNMSNRN-UZTKS`	`25`		
5	Check value `PBAGBFB-INHYG`	`13`		
6	30 documents from `create_document`, keys 1 to 25	Each returns its own key		

Done criteria

☐ Every test plan row returns exactly the expected key.